

Chapter 6.10

Provisions for the design, construction, inspection and testing of portable tanks with shells made of fibre-reinforced plastics (FRP) materials

6.10.1 Application and general requirements

- 6.10.1.1 The requirements of section 6.10.2 apply to portable tanks with an FRP shell intended for the transport of dangerous goods of classes or divisions 1, 3, 5.1, 6.1, 6.2, 8 and 9 by all modes of transport. In addition to the requirements of this chapter, unless otherwise specified, the applicable requirements of the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall be fulfilled by any multimodal portable tank with FRP shell which meets the definition of a “container” within the terms of that Convention.
- 6.10.1.2 The provisions of this chapter do not apply to offshore portable tanks.
- 6.10.1.3 The provisions of chapter 4.2 and section 6.7.2 apply to FRP portable tank shells except for those concerning the use of metal materials for the construction of a portable tank shell and additional provisions stated in this chapter.
- 6.10.1.4 In recognition of scientific and technological advances, the technical requirements of this chapter may be varied by alternative arrangements. These alternative arrangements shall offer a level of safety not less than that given by the provisions of this chapter with respect to compatibility with substances transported and the ability of the FRP portable tank to withstand impact, loading and fire conditions. For international transport, alternative arrangement FRP portable tanks shall be approved by the applicable competent authorities.

6.10.2 Provisions for the design, construction, inspection and testing of FRP portable tanks

6.10.2.1 Definitions

For the purposes of this section, the definitions of 6.7.2.1 apply except for definitions related to metal materials (“Fine grain steel”, “Mild steel” and “Reference steel”) for the construction of the shell of a portable tank.

Additionally, the following definitions apply to portable tanks with an FRP shell:

External layer means the part of the shell which is directly exposed to the atmosphere.

Fibre-reinforced plastic (FRP) means material consisting of fibrous and/or particulate reinforcement contained within a thermoset or thermoplastic polymer (matrix).

Filament winding means a process for constructing FRP structures in which continuous reinforcements (filament, tape, or other), either previously impregnated with a matrix material or impregnated during winding, are placed over a rotating mandrel. Generally, the shape is a surface of revolution and may include ends (heads).

FRP shell means a closed part of cylindrical shape with an interior volume intended for transport of chemical substances.

FRP tank means a portable tank constructed with an FRP shell and ends (heads), service equipment, safety relief devices and other installed equipment.

Glass transition temperature (T_g) means a characteristic value of the temperature range over which the glass transition takes place.

Hand layup means a process for moulding reinforced plastics in which reinforcement and resin are placed on a mould.

Liner means a layer on the inner surface of an FRP shell preventing contact with the dangerous goods being transported.

Part 6 – Construction and testing of packagings, IBCs, etc.

Mat means a fibre reinforcement made of random, chopped or twisted fibres bonded together as sheets of various length and thickness.

Parallel shell sample means an FRP specimen, which must be representative of the shell, constructed in parallel to the shell construction if it is not possible to use cut-outs from the shell itself. The parallel shell sample may be flat or curved.

Representative sample means a sample cut out from the shell.

Resin infusion means an FRP construction method by which dry reinforcement is placed into a matched mould, single-sided mould with vacuum bag, or otherwise, and liquid resin is supplied to the part through the use of external applied pressure at the inlet and/or application of full or partial vacuum pressure at the vent.

Structural layer means FRP layers of a shell required to sustain the design loads.

Veil means a thin mat with high absorbency used in FRP product plies where polymeric matrix surplus fraction content is required (surface evenness, chemical resistance, leakage-proof, etc.).

6.10.2.2 General design and construction provisions

6.10.2.2.1 The provisions of 6.7.1 and 6.7.2.2 apply to FRP portable tanks. For areas of the shell that are made from FRP, the following provisions of chapter 6.7 are exempt: 6.7.2.2.1, 6.7.2.2.9.1, 6.7.2.2.13 and 6.7.2.2.14. Shells shall be designed and constructed in accordance with the requirements of a pressure vessel code, applicable to FRP materials, recognized by the competent authority.

In addition, the following requirements apply.

6.10.2.2.2 Manufacturer's quality system

6.10.2.2.2.1 The quality system shall contain all the elements, requirements and provisions adopted by the manufacturer. It shall be documented in a systematic and orderly manner in the form of written policies, procedures and instructions.

6.10.2.2.2.2 The contents shall in particular include adequate descriptions of:

- .1 the organizational structure and responsibilities of personnel with regard to design and product quality;
- .2 the design control and design verification techniques, processes and procedures that will be used when designing the portable tanks;
- .3 the relevant manufacturing, quality control, quality assurance and process operation instructions that will be used;
- .4 quality records, such as inspection reports, test data and calibration data;
- .5 management reviews to ensure the effective operation of the quality system arising from the audits in accordance with 6.10.2.2.2.4;
- .6 the process describing how customer requirements are met;
- .7 the process for control of documents and their revision;
- .8 the means for control of non-conforming portable tanks, purchased components, in-process and final materials; and
- .9 the training programmes and qualification procedures for relevant personnel.

6.10.2.2.2.3 Under the quality system, the following minimum requirements shall be met for each FRP portable tank manufactured:

- .1 use of an inspection and test plan (ITP);
- .2 visual inspections;
- .3 verification of fibre orientation and mass fraction by means of a documented control process;
- .4 verification of fibre and resin quality and characteristics by means of certificates or other documentation;
- .5 verification of liner quality and characteristics by means of certificates or other documentation;
- .6 verification of whichever is applicable of formed thermoplastic resin characteristic or degree of cure of thermoset resin, by direct or indirect means (e.g. Barcol test or differential scanning calorimetry) to be determined in accordance with 6.10.2.7.1.2.8, or by creep testing of a representative sample or parallel shell specimen in accordance with 6.10.2.7.1.2.5 for a period of 100 hours;
- .7 documentation of whichever is applicable of thermoplastic resin forming processes or thermoset resin cure and post-cure processes; and
- .8 retention and archiving of shell samples for future inspection and shell verification (e.g. from manhole cut-out) for a period of five years.

6.10.2.2.2.4 Audit of the quality system

The quality system shall be initially assessed to determine whether it meets the provisions in 6.10.2.2.2.1 to 6.10.2.2.2.3 to the satisfaction of the competent authority.

The manufacturer shall be notified of the results of the audit. The notification shall contain the conclusions of the audit and any corrective actions required.

Periodic audits shall be carried out, to the satisfaction of the competent authority, to ensure that the manufacturer maintains and applies the quality system. Reports of the periodic audits shall be provided to the manufacturer.

6.10.2.2.2.5 Maintenance of the quality system

The manufacturer shall maintain the quality system as approved in order that it remains adequate and efficient.

The manufacturer shall notify the competent authority that approved the quality system of any intended changes. The proposed changes shall be evaluated to determine whether the amended quality system will still satisfy the provisions in 6.10.2.2.2.1 to 6.10.2.2.2.3.

6.10.2.2.3 FRP Shells

6.10.2.2.3.1 FRP shells shall have a secure connection with structural elements of the portable tank frame. FRP shell supports and attachments to the frame shall cause no local stress concentrations exceeding the design allowables of the shell structure in accordance with the provisions stated in this chapter for all operating and test conditions.

6.10.2.2.3.2 Shells shall be made of suitable materials, capable of operating within a minimum design temperature range of -40°C to $+50^{\circ}\text{C}$, unless temperature ranges are specified for specific more severe climatic or operating conditions (e.g. heating elements), by the competent authority of the country where the transport operation is being performed.

6.10.2.2.3.3 If a heating system is installed, it shall comply with 6.7.2.5.12 to 6.7.2.5.15 and with the following provisions:

- .1 the maximum operating temperature of the heating elements integrated or connected to the shell shall not exceed the maximum design temperature of the tank;
- .2 the heating elements shall be designed, controlled and utilized so that the temperature of the carried substance cannot exceed the maximum design temperature of the tank or a value at which the internal pressure exceeds MAWP; and
- .3 structures of the tank and its heating elements shall allow examination of the shell with respect to possible effects of overheating.

6.10.2.2.3.4 Shells shall consist of the following elements:

- liner;
- structural layer; and
- external layer.

Note: The elements may be combined if all applicable functional criteria are met.

6.10.2.2.3.5 The liner is the inner element of the shell designed as the primary barrier to provide for the long-term chemical resistance in relation to the substances to be carried, to prevent any dangerous reaction with the contents or the formation of dangerous compounds and any substantial weakening of the structural layer owing to the diffusion of products through the liner. Chemical compatibility shall be verified in accordance with 6.10.2.7.1.3.

The liner may be an FRP liner or a thermoplastic liner.

6.10.2.2.3.6 FRP liners shall consist of the following two components:

- .1 Surface layer ("gel-coat"): adequate resin rich surface layer, reinforced with a veil, compatible with the resin and contents. This layer shall have a maximum fibre mass content of 30% and have a minimum thickness of 0.25 mm and a maximum thickness of 0.60 mm.
- .2 Strengthening layer(s): layer or several layers with a minimum thickness of 2 mm, containing a minimum of 900 g/m^2 of glass mat or chopped fibres with a mass content in glass of not less than 30% unless equivalent safety is demonstrated for a lower glass content.

6.10.2.2.3.7 If the liner consists of thermoplastic sheets, they shall be welded together in the required shape, using a qualified welding procedure and personnel. Welded liners shall have a layer of electrically conductive media placed against the non-liquid contact surface of the welds to facilitate spark testing. Durable bonding between liners and the structural layer shall be achieved by the use of an appropriate method.

Part 6 – Construction and testing of packagings, IBCs, etc.

6.10.2.2.3.8 The structural layer shall be designed to withstand the design loads according to 6.7.2.2.12, 6.10.2.2.3.1, 6.10.2.3.2, 6.10.2.3.4 and 6.10.2.3.6.

6.10.2.2.3.9 The external layer of resin or paint shall provide adequate protection of the structural layers of the tank from environmental and service exposure, including UV radiation and salt fog, and occasional splash exposure to cargoes.

6.10.2.2.3.10 Resins

The processing of the resin mixture shall be carried out in compliance with the recommendations of the supplier. These resins can be:

- unsaturated polyester resins;
- vinyl ester resins;
- epoxy resins;
- phenolic resins; and
- thermoplastic resins.

The resin heat distortion temperature (HDT), determined in accordance with 6.10.2.7.1.1 shall be at least 20°C higher than the maximum design temperature of the shell as defined in 6.10.2.2.3.2, but shall in any case not be lower than 70°C.

6.10.2.2.3.11 Reinforcement material

The reinforcement material of the structural layers shall be selected such that they meet the requirements of the structural layer.

For the liner glass fibres of at a minimum type C or ECR according to ISO 2078:1993 + Amd 1:2015 shall be used. Thermoplastic veils may only be used for the liner when their compatibility with the intended contents has been demonstrated.

6.10.2.2.3.12 Additives

Additives necessary for the treatment of the resin, such as catalysts, accelerators, hardeners and thixotropic substances as well as materials used to improve the tank, such as fillers, colours, pigments, etc. shall not cause weakening of the material, taking into account lifetime and temperature expectancy of the design.

6.10.2.2.3.13 FRP shells, their attachments and their service and structural equipment shall be designed to withstand the loads mentioned in 6.7.2.2.12, 6.10.2.2.3, 6.10.2.3.2, 6.10.2.3.4 and 6.10.2.3.6 without loss of contents (other than quantities of gas escaping through any degassing vents) during the design lifetime.

6.10.2.2.3.14 Special provisions for the carriage of substances with a flashpoint of not more than 60°C

6.10.2.2.3.14.1 FRP tanks used for the carriage of flammable liquids with a flashpoint of not more than 60°C shall be constructed to ensure the elimination of static electricity from the various component parts to avoid the accumulation of dangerous charges.

6.10.2.2.3.14.2 The electrical surface resistance of the inside and outside of the shell as established by measurements shall not be higher than $10^9 \Omega$. This may be achieved by the use of additives in the resin or interlaminar conducting sheets, such as metal or carbon network.

6.10.2.2.3.14.3 The discharge resistance to earth as established by measurements shall not be higher than $10^7 \Omega$.

6.10.2.2.3.14.4 All components of the shell shall be electrically connected to each other and to the metal parts of the service and structural equipment of the tank and, when applicable, to the vehicle. The electrical resistance between components and equipment in contact with each other shall not exceed 10Ω .

6.10.2.2.3.14.5 The electrical surface-resistance and discharge resistance shall be measured initially on each manufactured tank or a specimen of the shell in accordance with the procedure recognized by the competent authority. In the event of damage to the shell, requiring repair, the electrical resistance shall be re-measured.

6.10.2.2.3.15 The tank shall be designed to withstand, without significant leakage, the effects of a full engulfment in fire for 30 minutes as specified by the test requirements in 6.10.2.7.1.5. Testing may be waived with the agreement of the competent authority, where sufficient proof can be provided by tests with comparable tank designs.

6.10.2.2.3.16 Construction process for FRP shells

6.10.2.2.3.16.1 Filament winding, hand layup, resin infusion or other appropriate composite production processes shall be used for construction of FRP shells.

Chapter 6.10 – Provisions for portable tanks with shells made of FRP materials

6.10.2.2.3.16.2 The weight of the fibre reinforcement shall conform to that set forth in the procedure specification with a tolerance of +10% and –0%. One or more of the fibre types specified in 6.10.2.2.3.11 and in the procedure specification shall be used for reinforcement of shells.

6.10.2.2.3.16.3 The resin system shall be one of the resin systems specified in 6.10.2.2.3.10. No filler, pigment, or dye additions shall be used which will interfere with the natural colour of the resin except as permitted by the procedure specification.

6.10.2.3 Design criteria

6.10.2.3.1 FRP shells shall be of a design capable of being stress-analysed mathematically or experimentally by resistance strain gauges, or by other methods approved by the competent authority.

6.10.2.3.2 FRP shells shall be designed and constructed to withstand the test pressure. Specific provisions are laid down for certain substances in the applicable portable tank instruction indicated in column 13 of the Dangerous Goods List and described in 4.2.5, or by a portable tank special provision indicated in column 14 of the Dangerous Goods List and described in 4.2.5.3. The minimum wall thickness of the FRP shell shall not be less than that specified in 6.10.2.4.

6.10.2.3.3 At the specified test pressure the maximum tensile relative deformation measured in mm/mm in the shell shall not result in the formation of microcracks, and therefore not be greater than the first measured point of elongation based fracture or damage of the resin, measured during tensile tests prescribed under 6.10.2.7.1.2.3.

6.10.2.3.4 For internal test pressure, external design pressure specified in 6.7.2.2.10, static loads specified in 6.7.2.2.12 and static gravity loads caused by the contents with the maximum density specified for the design and at maximum filling degree, failure criteria (FC) in the longitudinal direction, circumferential direction, and any other in-plane direction of the composite layout shall not exceed the following value:

$$FC \leq \frac{1}{K}$$

where:

$$K = K_0 \times K_1 \times K_2 \times K_3 \times K_4 \times K_5$$

where:

K shall have a minimum value of 4.

K_0 is a strength factor. For the general design the value for K_0 shall be equal to or more than 1.5. The value of K_0 shall be multiplied by a factor of two, unless the shell is provided with protection against damage consisting of a complete metal skeleton including longitudinal and transverse structural members.

K_1 is a factor related to the deterioration in the material properties due to creep and ageing. It shall be determined by the formula:

$$K_1 = \frac{1}{\alpha\beta}$$

where “ α ” is the creep factor and “ β ” is the ageing factor determined in accordance with 6.10.2.7.1.2.5 and .6, respectively. When used in calculation, factors α and β shall be between 0 and 1.

Alternatively, a conservative value of $K_1 = 2$ may be applied for the purpose of undertaking the numerical validation exercise in 6.10.2.3.4 (this does not remove the need to perform testing to determine α and β).

K_2 is a factor related to the service temperature and the thermal properties of the resin, determined by the following equation, with a minimum value of 1: $K_2 = 1.25 - 0.0125 (\text{HDT} - 70)$ where HDT is the heat distortion temperature of the resin, in °C.

K_3 is a factor related to the fatigue of the material; the value of $K_3 = 1.75$ shall be used unless otherwise agreed with the competent authority. For the dynamic design as outlined in 6.7.2.2.12 the value of $K_3 = 1.1$ shall be used.

K_4 is a factor related to resin curing and has the following values:

1.0 where curing is carried out in accordance with an approved and documented process, and the quality system described under 6.10.2.2.2 includes verification of degree of cure for each FRP portable tank using a direct measurement approach, such as differential scanning calorimetry (DSC) determined via ISO 11357-2:2016, as per 6.10.2.7.1.2.8.1.

1.1 where thermoplastic resin forming or thermoset resin curing is carried out in accordance with an approved and documented process, and the quality system described under 6.10.2.2.2 includes verification of whichever is applicable, formed thermoplastic resin characteristics or degree of cure of thermoset resin, for each FRP portable tank using an indirect measurement approach as per 6.10.2.7.1.2.8, such as Barcol testing via

Part 6 – Construction and testing of packagings, IBCs, etc.

ASTM D2583:2013-03 or EN 59:2016, HDT via ISO 75-1:2013, thermo-mechanical analysis (TMA) via ISO 11359-1:2014, or dynamic thermo-mechanical analysis (DMA) via ISO 6721-11:2019.

1.5 in other cases.

K_5 is a factor related to the portable tank instruction in 4.2.5.2.6:

1.0 for T1 to T19.

1.33 for T20.

1.67 for T21 to T22.

A design validation exercise using numerical analysis and a suitable composite failure criterion is to be undertaken to verify that the stresses in the plies in the shell are below the allowables. Suitable composite failure criteria include, but are not limited to, Tsai-Wu, Tsai-Hill, Hashin, Yamada-Sun, Strain Invariant Failure Theory, Maximum Strain or Maximum Stress. Other relations for the strength criteria are allowed upon agreement with the competent authority. The method and results of this design validation exercise are to be submitted to the competent authority.

The allowables are to be determined using experiments to derive parameters required by the chosen failure criteria combined with factor of safety K , the strength values measured as per 6.10.2.7.1.2.3, and the maximum elongation strain criteria prescribed in 6.10.2.3.5. The analysis of joints is to be undertaken in accordance with the allowables determined in 6.10.2.3.7 and the strength values measured as per 6.10.2.7.1.2.7. Buckling is to be considered in accordance with 6.10.2.3.6. Design of openings and metallic inclusions is to be considered in accordance with 6.10.2.3.8.

- 6.10.2.3.5** At any of the stresses as defined in 6.7.2.2.12 and 6.10.2.3.4, the resulting elongation in any direction shall not exceed the value indicated in the following table or one tenth of the elongation at fracture of the resin determined by ISO 527 2:2012, whichever is lower.

Examples of known limits are presented in the table below.

Type of resin	Maximum strain in tension (%)
Unsaturated polyester or phenolic	0.2
Vinylester	0.25
Epoxy	0.3
Thermoplastic	See 6.10.2.3.3

- 6.10.2.3.6** For the external design pressure, the minimum safety factor for linear buckling analysis of the shell shall be as defined in the applicable pressure vessel code but not less than three.

- 6.10.2.3.7** The adhesive bondlines and/or overlay laminates used in the joints, including the end joints, connection between the equipment and shell, the joints of the surge plates and the partitions with the shell shall be capable of withstanding the loads of 6.7.2.2.12, 6.10.2.2.3.1, 6.10.2.3.2, 6.10.2.3.4 and 6.10.2.3.6. In order to avoid concentrations of stresses in the overlay lamination, the applied taper shall not be steeper than 1:6. The shear strength between the overlay laminate and the tank components to which it is bonded shall not be less than:

$$\tau = y \frac{Q}{l} \leq \frac{\tau_R}{K}$$

where:

τ_R is the interlaminar shear strength according to ISO 14130:1997 and Cor 1:2003;

Q is the load per unit width of the interconnection;

K is the safety factor determined as per 6.10.2.3.4;

l is the length of the overlay laminate;

y is the notch factor relating average joint stress to peak joint stress at failure initiation location.

Other calculation methods for the joints are allowed following approval with the competent authority.

- 6.10.2.3.8** Metallic flanges and their closures are permitted to be used in FRP shells, under design provisions of 6.7.2. Openings in the FRP shell shall be reinforced to provide at least the same safety factors against the static and dynamic stresses as specified in 6.7.2.2.12, 6.10.2.3.2, 6.10.2.3.4 and 6.10.2.3.6 as that for the shell itself. The number of openings shall be minimized. The axis ratio of oval-shaped openings shall be not more than 2.

If metallic flanges or componentry are integrated into the FRP shell using bonding, then the characterization method stated in 6.10.2.3.7 shall apply to the joint between the metal and FRP. If the metallic flanges or componentry are fixed in an alternative fashion, e.g. threaded fastener connections, then the appropriate provisions of the relevant pressure vessel standard shall apply.

Chapter 6.10 – Provisions for portable tanks with shells made of FRP materials

- 6.10.2.3.9** Check calculations of the strength of the shell shall be performed by finite element method simulating the shell layups, joints within FRP shell, joints between the FRP shell and the container frame, and openings. Treatment of singularities shall be undertaken using an appropriate method according to the applicable pressure vessel code.
- 6.10.2.4** **Minimum wall thickness of the shell**
- 6.10.2.4.1** Minimum thickness of the FRP shell shall be confirmed by check calculations of the strength of the shell considering strength provisions given in 6.10.2.3.4.
- 6.10.2.4.2** Minimum thickness of the FRP shell structural layers shall be determined in accordance with 6.10.2.3.4; however, in any case the minimum thickness of the structural layers shall be at least 3 mm.
- 6.10.2.5** **Equipment components for portable tanks with FRP shell**
- Service equipment, bottom openings, pressure relief devices, gauging devices, supports, frameworks, lifting and tie-down attachments of portable tanks shall meet the provisions of 6.7.2.5 to 6.7.2.17. If any other metallic features are required to be integrated into the FRP shell, then the provisions of 6.10.2.3.8 shall apply.
- 6.10.2.6** **Design approval**
- 6.10.2.6.1** Design approval of FRP portable tanks shall be as per 6.7.2.18 provisions. The following additional provisions apply to FRP portable tanks.
- 6.10.2.6.2** The prototype test report for the purpose of the design approval shall additionally include the following:
- .1 results of the material tests used for FRP shell fabrication in accordance with 6.10.2.7.1 provisions;
 - .2 results of the ball drop test in accordance with the provisions of 6.10.2.7.1.4; and
 - .3 results of the fire resistance test in accordance with the provisions of 6.10.2.7.1.5.
- 6.10.2.6.3** A service life inspection programme shall be established, which shall be a part of the operation manual, to monitor the condition of the tank at periodic inspections. The inspection programme shall focus on the critical stress locations identified in the design analysis performed under 6.10.2.3.4. The inspection method shall take into account the potential damage mode at the critical stress location (e.g. tensile stress or interlaminar stress). The inspection shall be a combination of visual and non-destructive testing (e.g. acoustic emissions, ultrasonic evaluation, thermographic). For heating elements, the service life inspection programme shall allow an examination of the shell or its representative locations to take into account the effects of overheating.
- 6.10.2.6.4** A representative prototype tank shall be subjected to tests as specified below. For this purpose, service equipment may be replaced by other items if necessary.
- 6.10.2.6.4.1** The prototype shall be inspected for compliance with the design type specification. This shall include an internal and external inspection and measurement of the main dimensions.
- 6.10.2.6.4.2** The prototype, equipped with strain gauges at all locations of high strain, as identified by the design validation exercise in accordance with 6.10.2.3.4, shall be subjected to the following loads and the strain shall be recorded:
- .1 Filled with water to the maximum filling degree. The measuring results shall be used to calibrate the design calculations according to 6.10.2.3.4.
 - .2 Filled with water to the maximum filling degree and subjected to static loads in all three directions mounted by the base corner castings without additional mass applied external to the shell. For comparison with the design calculation according to 6.10.2.3.4 the strains recorded shall be extrapolated in relation to the quotient of the accelerations required in 6.7.2.2.12 and measured.
 - .3 Filled with water and subjected to the specified test pressure. Under this load, the shell shall exhibit no visual damage or leakage. The stress corresponding to the measured strain level shall not exceed the minimum factor of safety calculated in 6.10.2.3.4 under any of these loading conditions.
- 6.10.2.7** **Additional provisions applicable to FRP portable tanks**
- 6.10.2.7.1** **Material testing**
- 6.10.2.7.1.1** *Resins*
- Resin tensile elongation shall be determined in accordance with ISO 527-2:2012. The heat distortion temperature (HDT) of the resin shall be determined in accordance with ISO 75-1:2013.

Part 6 – Construction and testing of packagings, IBCs, etc.**6.10.2.7.1.2 Shell samples**

Prior to testing, all coatings shall be removed from the samples. If shell samples are not possible then parallel shell samples may be used. The tests shall cover:

- .1 Thickness of the laminates of the central shell wall and the ends.
- .2 Mass content and composition of composite reinforcement by ISO 1172:1996 or ISO 14127:2008, as well as orientation and arrangement of reinforcement layers.
- .3 Tensile strength, elongation at fracture and modulus of elasticity according to ISO 527-4:1997 or ISO 527-5:2009 for the circumferential and longitudinal directions of the shell. For areas of the FRP shell, tests shall be performed on representative laminates in accordance with ISO 527-4:1997 or ISO 527-5:2009, to permit evaluation of the suitability of safety factor (K). A minimum of six specimens per measure of tensile strength shall be used, and the tensile strength shall be taken as the average minus two standard deviations.
- .4 Bending deflection and strength shall be established by the three-point or four-point bending test according to ISO 14125:1998 + Amd 1:2011 using a sample with a minimum width of 50 mm and a support distance of at least 20 times the wall thickness. A minimum of five specimens shall be used.
- .5 Creep factor α shall be determined by taking the average result of at least two specimens with the configuration described in .4, subject to creep in three-point or four-point bending, at the maximum design temperature nominated under 6.10.2.2.3.2, for a period of 1,000 hours. The following test is to be undertaken for each specimen:
 - .1 Place specimen into bending apparatus, unloaded, in oven set to maximum design temperature and allow to acclimatize for a period of not less than 60 minutes.
 - .2 Load specimen bending in accordance with ISO 14125:1998 + Amd 1:2011 at flexural stress equal to the strength determined in .4 divided by four. Maintain mechanical load at maximum design temperature without interruption for not less than 1,000 hours;
 - .3 Measure the initial deflection six minutes after full load application in sub-paragraph .2 above. Specimen shall remain loaded in test rig.
 - .4 Measure the final deflection 1,000 hours after full load application in sub-paragraph .2 above.
 - .5 Calculate the creep factor α by dividing the initial deflection from sub-paragraph .3 above by the final deflection from sub-paragraph .4 above.
- .6 Ageing factor β shall be determined by taking the average result of at least two specimens with the configuration described in .4, subject to loading in static three-point or four-point bending, in conjunction with immersion in water at the maximum design temperature nominated under 6.10.2.2.3.2 for a period of 1,000 hours. The following test is to be undertaken for each specimen:
 - .1 Prior to testing or conditioning, specimens shall be dried in an oven at 80°C for a period of 24 hours.
 - .2 The specimen shall be loaded in three-point or four-point bending at ambient temperature, in accordance with ISO 14125:1998 + Amd 1:2011, at the flexural stress level equal to the strength determined in .4 divided by four. Measure the initial deflection six minutes after full load application. Remove specimen from test rig.
 - .3 Immerse unloaded specimen in water at the maximum design temperature for a period of not less than 1,000 hours without interruption to the water conditioning period. When the conditioning period has lapsed, remove specimens, keep damp at ambient temperature, and complete sub-paragraph .4 below within three days.
 - .4 The specimen shall be subject to a second round of static loading, in a manner identical to sub-paragraph .2 above. Measure the final deflection six minutes after full load application. Remove specimen from test rig.
 - .5 Calculate the ageing factor β by dividing the initial deflection from sub-paragraph .2 above by the final deflection from sub-paragraph .4.
- .7 The interlaminar shear strength of the joints shall be measured by testing representative samples in accordance with ISO 14130:1997.
- .8 The efficiency of whichever is applicable of thermoplastic resin forming characteristics or thermoset resin cure and post-cure processes for laminates are to be determined using one or more of the following methods:
 - .1 direct measurement of formed thermoplastic resin characteristics or thermoset resin degree of cure: glass transition temperature (T_g) or melting temperature (T_m) determined using differential scanning calorimetry (DSC) via ISO 11357-2:2016; or
 - .2 indirect measurement of formed thermoplastic resin characteristics or thermoset resin degree of cure:
 - HDT via ISO 75-1:2013;
 - T_g or T_m using thermo-mechanical analysis (TMA) via ISO 11359-1:2014;
 - dynamic thermo-mechanical analysis (DMA) via ISO 6721-11:2019;
 - Barcol testing via ASTM D2583:2013-03 or EN 59:2016.

Chapter 6.10 – Provisions for portable tanks with shells made of FRP materials

6.10.2.7.1.3 The chemical compatibility of the liner and chemical contact surfaces of service equipment with the substances to be carried shall be demonstrated by one of the following methods. This demonstration shall account for all aspects of the compatibility of the materials of the shell and its equipment with the substances to be carried, including chemical deterioration of the shell, initiation of critical reactions of the contents and dangerous reactions between both.

- .1 In order to establish any deterioration of the shell, representative samples taken from the shell, including any liners with welds, shall be subjected to the chemical compatibility test according to EN 977:1997 for a period of 1,000 hours at 50°C or the maximum temperature at which a particular substance is approved for transport. Compared with a virgin sample, the loss of strength and elasticity modulus measured by the bending test according to EN 978:1997 shall not exceed 25%. Cracks, bubbles, pitting effects as well as separation of layers and liners and roughness shall not be acceptable.
- .2 Certified and documented data of positive experiences on the compatibility of filling substances in question with the materials of the shell with which they come into contact at given temperatures, times and other relevant service conditions.
- .3 Technical data published in relevant literature, standards or other sources, acceptable to the competent authority.
- .4 Upon agreement with the competent authority other methods of chemical compatibility verification may be used.

6.10.2.7.1.4 *Ball drop test as per EN 976-1:1997*

The prototype shall be subjected to the ball drop test according to EN 976-1:1997, No. 6.6. No visible damage inside or outside the tank shall occur.

6.10.2.7.1.5 *Fire resistance test*

6.10.2.7.1.5.1 A representative prototype tank with its service and structural equipment in place and filled to 80% of its maximum capacity with water, shall be exposed to a full engulfment in fire for 30 minutes, caused by an open heating oil pool fire or any other type of fire with the same effect. The fire shall be equivalent to a theoretical fire with a flame temperature of 800°C, emissivity of 0.9 and to the tank a heat transfer coefficient of 10 W/(m²K) and surface absorptivity of 0.8. A minimum net heat flux of 75 kW/m² shall be calibrated according to ISO 21843:2018. The dimensions of the pool shall exceed those of the tank by at least 50 cm to each side and the distance between fuel level and tank shall be between 50 cm and 80 cm. The rest of the tank below liquid level, including openings and closures, shall remain leakproof except for drips.

6.10.2.8 **Inspection and testing**

6.10.2.8.1 Inspection and testing of portable FRP tanks shall be carried out as per provisions of 6.7.2.19. In addition, welded thermoplastic liners shall be spark tested under a suitable standard, after pressure tests performed in accordance with the periodic inspections specified in 6.7.2.19.4.

6.10.2.8.2 In addition, the initial and periodic inspections shall follow the service life inspection programme and any associated inspection methods per 6.10.2.6.3.

6.10.2.8.3 The initial inspection and test shall verify that construction of the tank is made in accordance with the quality system required by 6.10.2.2.2.

6.10.2.8.4 Additionally, during inspection of the shell the position of the areas heated by heating elements shall be indicated or marked, be available on design drawings or shall be made visible by a suitable technique (e.g. infrared). Examination of the shell shall take into account the effects of overheating, corrosion, erosion, overpressure and mechanical overloading.

6.10.2.9 **Retention of samples**

Shell samples (e.g. from manhole cut-out) for each tank manufactured shall be maintained for future inspection and shell verification for a period of five years from the date of the initial inspection and test and until successful completion of the required 5-year periodic inspection.

6.10.2.10 **Marking**

6.10.2.10.1 The requirements of 6.7.2.20.1 apply to portable tanks with an FRP shell except those of 6.7.2.20.1(f)(ii).

6.10.2.10.2 The information required in 6.7.2.20.1(f)(i) shall be "Shell structural material: Fibre-reinforced plastic", the reinforcement fibre e.g. "Reinforcement: E-glass", and resin e.g. "Resin: Vinyl Ester".

6.10.2.10.3 Requirements of provision 6.7.2.20.2 apply to portable tank with an FRP shell.